

▼ Pyramid Problems · Inverted Pyramid — English Worksheet

Topic: Inverted Pyramid · **Course:** Pyramid Problems · **Time:** about 45 minutes

1 · Predict 🧠

Read each set of steps. Write what you think will happen.

```
set size to 1
while size <= 9:
    [build one size × size square layer]
    move up by 1
    set size to size + 2
```

Each layer gets bigger or smaller? Where is the widest layer — bottom or top?


```
set size to 1
while size <= 9:
    [build one size × size square layer]
    move up by 1
    set size to size + 2
```

List the sizes from bottom to top. How many layers in total?


```
set size to 1
while size <= 5:
    [build one size × size square layer]
    move up by 1
    set size to size + 2
```

How is this different from the code above? How does it look?

2 · Spot the Bug 🐛

Each code block below is broken. Rewrite it so it works, then explain the fix.

Bug A — The agent is supposed to build an **inverted pyramid**: the smallest layer at the bottom, the biggest at the top.

```
set size to 9
while size > 0:
    [build one size × size square layer]
    move up by 1
    set size to size - 2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The agent is supposed to grow the pyramid from 1 up to 9. Each layer must be **2 blocks longer** than the one below.

```
set size to 1
while size <= 9:
    [build one size × size square layer]
    move up by 1
    set size to size + 1
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Each new layer should sit **on top** of the one below.

```
set size to 1
while size <= 9:
    [build one size × size square layer]
    set size to size + 2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world. Build an inverted pyramid that grows from **size 1 up to size 9**, then check the shape from the side.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.